

Recitation 03

Variables, User Input & Output

Question 1 | Variable assignment

Consider the following variable assignments:

```
Left = "Lady"
Middle = "Jack"
Right = "Ace"
```

多重赋值

1) Figure out where the 'Lady' is (Left, Middle, Right) after running the following programs:

	Left, Right, Middle = Left, Middle, Right	
Left ✓	Left	lady
middle	Left, Right, Middle = Middle, Right, Left	
	Right middle	lady
Right	Left, Middle, Right = Middle, Right, Left	
	Right	
middle	Left, Middle, Right = Middle, Right, Left	
	Left, Middle, Right = Middle, Right, Left	
	Middle	"Right" "left" "middle"
Left ✓	Left, Middle, Right = Middle, Right, Left	
	Left, Middle, Right = Middle, Right, Left	
	Left, Middle, Right = Middle, Right, Left	
	Middle Left	
	Left, Middle, Right = Right, Right, Right	
	Left, Middle, Right = Left, Left, Left	
	Left, Middle, Right = Middle, Middle, Middle	
NO.	NO	

① 先读取右边的值
② 平移赋值给左边
③ 可以连续赋值

Question 2 | Variable swap

Consider the following variable assignments:

```
a = 1
b = 2
c = a
```

Handwritten notes: = 1 → 此时就赋值好了, 不可改变

1) Figure out the output of the following statements:

<pre>print(a, b, c)</pre>	1 2 1
<pre>a = b print(a, b, c)</pre>	2 2 2 2 2 1 <i>因为c=1 此时就赋值好了</i>
<pre>b = c print(a, b, c)</pre>	1 1 1

111

Question 3 | Print arguments

<pre>a = 5 print("a", a, sep=' = ')</pre>	"5" = 5
<pre>a = 5 print("a = ", a, sep="0000")</pre>	a = 0000 5
<pre>a = 5 print("a = ", a, sep="'", end="''")</pre>	a = ' 5'
<pre>a, b = 'b', 'a' print(a, 'b', sep='>>>', end='\t') print('a', b, sep='***', end='\n\n') print(a, b, sep='\t')</pre>	'b' >>> 'b' [] 'a' *** 'a' \n 'b' [] 'a' <i>Handwritten notes:</i> ↙ 引号去掉 ↘ 不换行 ↖ b >>> b ↗ 换 b a a *** a

Question 4 | String formatting

<pre>print("{:8d}".format(12))</pre>	<i>用空格</i> "0000012" "12"
<pre>print("{:0>8}".format('12'))</pre>	"00000012"
<pre>print("{:8.3f}".format(12.34567890))</pre>	<i>没说明</i> "00012.346"
<pre>print("{:08.3f}".format(12.34567890))</pre>	<i>也算一格</i> "00012.346"
<pre>print("{:^08.3f}".format(12.34567890))</pre>	"012.346000"
<pre>print("{:<08d}".format(12))</pre>	"12000000"
<pre>{:>9}'.format('cat')</pre>	<i>空格</i> "00000000cat"
<pre>'{: *^9}'.format('cat')</pre>	"***cat***"
<pre>'{: .3}'.format('caterpillar')</pre>	"cat"
<pre>'{:>9.3}'.format('caterpillar')</pre>	<i>空格</i> "000000000cat"
<pre>'{: -^9.3}'.format('caterpillar')</pre>	"--cat--"


```

projected_sales = intfloat(input("Your projected amount of total sales: "))
annual_profit = 0.23 * projected_sales
print("annual-profit")

```

Question 5 | Sales Prediction

A company has determined that its annual profit is typically 23 percent of total sales.

- 1) Write a program that asks the user to enter the projected amount of total sales, and then displays the profit that will be made from that amount.

一般是浮点数

Question 6 | Ingredient Adjuster

A cookie recipe calls for the following ingredients:

- 1.5 cups of sugar
- 1 cup of butter
- 2.75 cups of flour

The recipe produces 48 cookies with this amount of the ingredients.

- 1) Write a program that asks the user how many cookies he or she wants to make, and then displays the number of cups of each ingredient needed for the specified number of cookies.

```

number = int(input("how many: "))
ratio = number / 48
sugar = 1.5 * ratio
butter = 1 * ratio
flour = 2.75 * ratio

```

```

print("You need {} cups of sugar, {} cups of butter
and {} cups of flour".format(sugar, butter, flour))

```